

Naeheon Lee

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EDUCATION

Portland State University	Portland, OR
<i>Master of Science in Computer Science</i>	<i>Expected February 2027</i>
Relevant Coursework: Intro to DBMS, Robust SW System Design, Data with Python, Algorithm Design/Analysis	
Kangwon National University	Samcheok, South Korea
<i>Bachelor of Engineering, Double Major: Computer Engineering and Data Science</i>	<i>February 2024</i>
Honors: Merit Based Scholarship (2018), National Program of Excellence in Software (2020, 2021)	

TECHNICAL SKILLS

- Programming Languages: Python (Proficient), C/C++, SQL, Java, Javascript.
- Web & Cloud: Django, Node.js, Streamlit, GitHub, AWS, GCP.
- Data Science & Machine Learning: Pandas, Numpy, Matplotlib, Seaborn, Plotly, Scikit-learn, SciPy.
- Core Competencies: Machine Learning, Data Modeling, ETL Pipelines, Software Design, Creative Problem Solving.

DATA SCIENCE & WEB PROJECTS

Jazz Discography Relational Database	Fall 2025
- Engineered a Python ETL script parsing a 30GB+ XML dump to extract a 300MB targeted jazz dataset. - Designed a 3NF SQLite schema and advanced SQL queries to analyze jazz subgenre evolution and historical trends.	
Portland Crime & Economic Trend Analysis	Fall 2025
- Engineered a Pandas ETL pipeline processing 650,000+ rows of crime data to analyze economic recession correlations. - Co-developed interactive Streamlit heatmaps with a 3-person team, accelerating web-based data visualization.	
Scalable Event Notification System	Fall 2025
- Architected a distributed service using Apache Kafka for high-throughput ingestion, employing horizontal scaling to handle thousands of events/sec. - Designed fault-tolerant Circuit Breakers and Distributed Caching to theoretically prevent cascading system failures.	
Virtual Fitting Room Web Application	Summer 2021
- Developed a virtual fitting room web app enabling users to dynamically try on 200+ clothing items via custom avatars. - Integrated an open-source background removal model, achieving under 2 second image processing latency.	

RESEARCH & PROFESSIONAL EXPERIENCE

Machine Learning Lab, Kangwon National University	Chuncheon, South Korea
<i>Undergraduate Research Assistant</i>	<i>June 2021 - August 2021</i>
- Co-engineered an arrhythmia prediction model with 5 researchers, increasing diagnostic accuracy from ~60% to 86%. - Designed a secure data acquisition framework to collect pseudonymized healthcare records from 3+ public institutions.	
Forestry and SW Engineering Convergence Unit Program	Chuncheon, South Korea
<i>Undergraduate Researcher</i>	<i>July 2020 - August 2021</i>
- Developed a Python analysis tool processing 700+ datasets to track <i>Scopolia japonica</i> habitat migration. - Built a preprocessed 15,000+ plant species database, utilizing Pearson's correlation to statistically validate distribution shifts.	
Kangwon National University Dept. of IT	Samcheok, South Korea
<i>Student IT Infrastructure Specialist</i>	<i>Mar 2019 – Mar 2020</i>
- Diagnosed and resolved hardware/software issues across 1,300+ computers, reducing overall system downtime by 20%. - Created a comprehensive computer maintenance handbook, decreasing routine troubleshooting time by 30%.	